Roll No. | | | | | | | |

EVEN END SEMESTER EXAMINATION - 2025

***Name of the Course**: B.Tech Aerospace Engineering* ***Semester:** Fourth*
***Name of the Paper:** Aerospace Materials and Manufacturing Technology* ***Paper Code:** TAS406*
***Time:** 3 Hours* ***Maximum Marks:** 100*

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	**(20 marks)**	**CO 1**
(a)	What is aerospace materials? What makes for a good aerospace material?	
(b)	What is solid solution? Differentiate substitional and interstitial solid solution with examples?	
(c)	Many aluminium alloys for aerospace applications are precipitation hardened. What is a precipitate and how does it make the aluminium alloys harder?	
Q2	**(20 marks)**	**CO2, CO3**
(a)	Explain following: Tension Strength, Compression Strength, Torsion Strength and Flexure Strength.	
(b)	Draw S-N diagram (schematic) for mild steel and for Aluminium. Explain how S-N diagram is plotted.	
(c)	What is the difference between Izod and Charpy impact testing methods? Explain briefly.	
Q3	**(20 marks)**	**CO3**
(a)	What is the difference between open-die and impression-die drop forging? Why is open-die forging not practical for large scale production of identical products? Why are multiple steps often required in the production of an impression-die forging?	
(b)	Explain the importance and function of gates, risers and runners in a mould. What are risers? Why is it desirable to restrict the size and use of risers?	
(c)	Differentiate among autogenous, homogeneous & heterogeneous welding processes.	
Q4	**(20 marks)**	**CO4**
(a)	Why are coolants required in metal cutting? What is the effect of a coolant on surface finish?	
(b)	Explain the basic steps in Powder Metallurgy.	
(c)	What do you mean by tool wear? Explain with neat sketches.	
Q5	**(20 marks)**	**CO5**
(a)	Define the term polymerization. What do you mean by copolymer and give two examples?	
(b)	What are ceramic materials? Differentiate between traditional and advanced ceramics.	
(c)	What are composite materials? What are the aircraft structural components made of composite materials?	